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Inventor(s)

Church; Mark et al.

SYSTEM, METHOD, AND COMPUTER DEVICE FOR DATA INTEGRITY VALIDATION

Abstract

A system, method, and device for data integrity validation are provided. The system includes a database for storing user data, a black box device for locally answering queries received from an end user device without revealing the stored user data, and end user device. The black box device includes memory for storing the user data, data filters, filtered data, the queries received from the end user device, query results, and excluded query result data and processor in communication with the memory. Processor is configured to generate the filtered data, determine whether each query received that addresses the filtered data is permitted, provide a response to each permitted query based on the generated filtered data, generate the excluded query result data by excluding stored user data that would facilitate identifying any user from the response to each permitted query, and transmit the excluded query result data to the end user device.

Inventors: Church; Mark (Waterloo, CA), Packull; Zachary (Cambridge, CA)

Applicant: Elpis Technologies Inc. (Kitchener, CA)

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Background/Summary

TECHNICAL FIELD

[0001] The following relates generally to data integrity validation, and more particularly to systems, methods, and devices for allowing users to control access to their data across services and platforms.

INTRODUCTION

[0002] A common concern of users and clients of various electronic and Internet-based services and platforms is a lack of control over the privacy and security of user data. Where user data may be sold, transmitted, repackaged, and reused without user consent, it is often difficult if not impossible to safeguard one's digital anonymity and ensure sensitive personal information is not leaked. Even where users provide consent, such consent is often in the form of a blanket agreement in order to gain access to a service or platform, and so users still lack control over their own data. Oftentimes when a user will not grant such consent, it becomes impossible for the user to fully participate in the digital economy and infrastructure underpinning most modern electronic services and platforms.

[0003] Accordingly, there is a need for allowing users to control their data. Specifically, there is a need for allowing users to meaningfully permit certain uses of user data and/or to prohibit entities from using certain user data in certain ways. There is a further need for allowing users to know where and to whom their data is being transmitted and how that data is to be used.

[0004] Accordingly, there is a need for preventing unauthorized, unwanted, and unwarranted copying and subsequent transmission of user data.

SUMMARY

[0005] A system for data integrity validation is provided. The system includes a user database for storing user data of a user and a black box device for locally answering queries received from an end user device without revealing the stored user data. The black box device includes a memory for storing the user data, data filters, filtered data, the queries received from the end user device, results of the queries, and excluded query result data and a processor in communication with the memory. The processor is configured to generate the filtered data based on the stored user data and the data filters, determine whether each query received that addresses the filtered data is permitted based on properties of the query, determine whether the query, if answered locally, would produce a result that would facilitate identifying any particular user, provide a response to each permitted query based on the generated filtered data, generate the excluded query result data by excluding from the response to each permitted query non-permitted stored user data and stored user data that would facilitate identifying any particular user, and transmit the excluded query result data to the end user device. The system further includes the end user device configured to receive the queries from an end user, transmit the queries to the black box device, and display the excluded query result data to the end user.

[0006] The black box device may include a permissions matrix for storing circumstances under which the user allows the user data to be used to respond to the queries, and the user data is not used to respond to the queries unless circumstances provided by the end user device are included in the permissions matrix.

[0007] The permissions matrix may notify the user when the circumstances provided by the end user device are not included in the permissions matrix. The permissions matrix may be configured

to receive permission from the user to allow the user data to be used to respond to the queries when the circumstances provided by the end user device are included in the permissions matrix.

[0008] Use by the end user of the excluded query result data may be audited to confirm that the use by the end user matches the circumstances provided by the end user device.

[0009] The black box device may further include a minimum bid amount for which the user allows the user data to be used to respond to the queries, and the user data is not used to respond to the queries unless a bid by the end user device meets or exceeds the minimum bid amount.

[0010] The minimum bid amount may include a first minimum bid amount for anonymous disclosure of the user data, a second minimum bid amount for semi-anonymous disclosure of the user data, the second minimum bid amount being higher than the first minimum bid amount, and a third minimum bid amount for non-anonymous disclosure of the user data, the third minimum bid amount being higher than the second minimum bid amount.

[0011] The system may further include a user device for providing the user data.

[0012] The user data may be gathered from any one or more of online activities of the user, purchase patterns of the user, and physical movement of the user.

[0013] A method for facilitating data integrity validation is provided. The method includes receiving user data pertaining to one or more users, receiving data filters to be applied to the user data, applying the data filters to the user data to generate filtered data, receiving a query from an end user device, determining whether the query received is permitted based on properties of the query, determining whether the query, if answered locally, would produce a result that would facilitate identifying any particular user, providing a response to the permitted query using the filtered data, generating excluded query result data by excluding from the response to each permitted query non-permitted user data and user data that would facilitate identifying any particular user, and transmitting the excluded query result data to the end user.

[0014] The method may further include storing circumstances under which the user allows the user data to be used to respond to the query in a permissions matrix and not allowing the user data to be used to respond to the query unless circumstances provided by the end user are included in the permissions matrix.

[0015] The permissions matrix may notify the user when the circumstances provided by the end user device are not included in the permissions matrix. The permissions matrix may receive permission from the user to allow the user data to be used to respond to the queries when the circumstances provided by the end user device are included in the permissions matrix.

[0016] The method may further include auditing use by the end user of the excluded query result data to confirm that the use by the end user matches the circumstances provided by the end user device.

[0017] The method may further include allowing the user data to be used to respond to the queries when a bid by the end user exceeds a minimum bid amount and not allowing the user data to be used to respond to the query if the bid by the end user device does not meet or exceed the minimum bid amount.

[0018] The minimum bid amount may include a first minimum bid amount for anonymous disclosure of the user data, a second minimum bid amount for semi-anonymous disclosure of the user data, the second minimum bid amount being higher than the first minimum bid amount, and a third minimum bid amount for non-anonymous disclosure of the user data, the third minimum bid amount being higher than the second minimum bid amount.

[0019] A black box device for locally answering queries received from an end user device without revealing stored user data of a user is provided. The black box device includes a memory for storing the user data, data filters, filtered data, the queries received from the end user device, results of the queries, and excluded query result data and a processor in communication with the memory. The processor is configured to generate the filtered data based on the stored user data and the data filters, determine whether each query received that addresses the filtered data is permitted based on

properties of the query, determine whether the query, if answered locally, would produce a result that would facilitate identifying any particular user, provide a response to each permitted query based on the generated filtered data, generate the excluded query result data by excluding from the response to each permitted query non-permitted stored user data and stored user data that would facilitate identifying any particular user, and transmit the excluded query result data to the end user device.

[0020] The black box device may further include a permissions matrix for storing circumstances under which the user allows the user data to be used to respond to the queries, and the user data is not used to respond to the queries unless circumstances provided by the end user are included in the permissions matrix.

[0021] The permissions matrix may notify the user when the circumstances provided by the end user device are not included in the permissions matrix. The permissions matrix may be configured to receive permission from the user to allow the user data to be used to respond to the queries when the circumstances provided by the end user device are included in the permissions matrix.

[0022] Use by the end user of the excluded query result data may be audited to confirm that the use by the end user matches the circumstances provided by the end user device.

[0023] The black box device may further include a minimum bid amount for which the user allows the user data to be used to respond to the queries, and the user data is not used to respond to the queries unless a bid by the end user device meets or exceeds the minimum bid amount.

[0024] The minimum bid amount may include a first minimum bid amount for anonymous disclosure of the user data, a second minimum bid amount for semi-anonymous disclosure of the user data, the second minimum bid amount being higher than the first minimum bid amount, and a third minimum bid amount for non-anonymous disclosure of the user data, the third minimum bid amount being higher than the second minimum bid amount.

[0025] Other aspects and features will become apparent, to those ordinarily skilled in the art, upon review of the following description of some exemplary embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The drawings included herewith are for illustrating various examples of articles, methods, and apparatuses of the present specification. In the drawings:

[0027] FIG. 1 is a schematic diagram of a system for data integrity validation, according to an embodiment;

[0028] FIG. 2 is a block diagram of a computing device of the present disclosure, according to an embodiment;

[0029] FIG. 3 is a block diagram of the system for data integrity validation of FIG. 1, according to an embodiment;

[0030] FIG. 4 is a flow diagram of a method of drawing inferences from user data without copying the user data, according to an embodiment;

[0031] FIG. 5 is a flow diagram of a method of granting users control over who may use their data and how that data may be used, according to an embodiment; and

[0032] FIG. 6 is a flow diagram of a method of using data marketplace pricing to facilitate user-directed transactions involving user data, according to an embodiment.

DETAILED DESCRIPTION

[0033] Various apparatuses or processes will be described below to provide an example of each claimed embodiment. No embodiment described below limits any claimed embodiment and any claimed embodiment may cover processes or apparatuses that differ from those described below. The claimed embodiments are not limited to apparatuses or processes having all of the features of

any one apparatus or process described below or to features common to multiple or all of the apparatuses described below.

[0034] One or more systems described herein may be implemented in computer programs executing on programmable computers, each comprising at least one processor, a data storage system (including volatile and non-volatile memory and/or storage elements), at least one input device, and at least one output device. For example, and without limitation, the programmable computer may be a programmable logic unit, a mainframe computer, server, and personal computer, cloud-based program or system, laptop, personal digital assistant, cellular telephone, smartphone, or tablet device.

[0035] Each program is preferably implemented in a high-level procedural or object-oriented programming and/or scripting language to communicate with a computer system. However, the programs can be implemented in assembly or machine language, if desired. In any case, the language may be a compiled or interpreted language. Each such computer program is preferably stored on a storage media or a device readable by a general or special purpose programmable computer for configuring and operating the computer when the storage media or device is read by the computer to perform the procedures described herein.

[0036] A description of an embodiment with several components in communication with each other does not imply that all such components are required. On the contrary, a variety of optional components are described to illustrate the wide variety of possible embodiments of the present invention.

[0037] Further, although process steps, method steps, algorithms or the like may be described (in the disclosure and/or in the claims) in a sequential order, such processes, methods and algorithms may be configured to work in alternate orders. In other words, any sequence or order of steps that may be described does not necessarily indicate a requirement that the steps be performed in that order. The steps of processes described herein may be performed in any order that is practical. Further, some steps may be performed simultaneously.

[0038] When a single device or article is described herein, it will be readily apparent that more than one device/article (whether or not they cooperate) may be used in place of a single device/article. Similarly, where more than one device or article is described herein (whether or not they cooperate), it will be readily apparent that a single device/article may be used in place of the more than one device or article.

[0039] The following relates generally to data integrity validation, and more particularly to systems, methods, and devices for allowing users to control access to their data across services and platforms. The present disclosure provides systems, methods, and devices for user-directed data integrity validation using black box devices and permission controls. The black box devices present in the system record data provided by users or otherwise gleaned from networks. However, according to permissions provided or otherwise agreed to by users, the data in the black box devices may only be accessed in certain ways.

[0040] In an embodiment, where third parties wish to access the data (for example, for marketing purposes), those third parties may not directly gain access to the data but may instead submit queries to the black box device to be performed and/or answered entirely locally. For example, where an advertiser wishes to know the locations of users who have recently performed a search for certain products and/or services, the advertiser submits a query to the black box device, which reviews the data stored internally and provides an answer (such as ‘50% of those users who searched for your product in Ontario live within the Greater Toronto Area’) that does not implicate specific user data. As a further example, the black box device may not respond to a query that implicates specific user data (such as ‘On what street does the person with the highest income who has searched for our product live?’).

[0041] In an embodiment, the black box device excludes or holds back parts of responses that implicate specific user data while still providing the rest of a query response that does not.

[0042] Furthermore, users who provide data to the system herein disclosed may dynamically set permissions for third parties to have access to the user data. Such access may be recorded in a log, such as a metadata log, available to the user. Such third parties may be required to specify the use to which the user data (or answers to queries associated with the user data as previously described) will be put. The black box may verify the use after the user data is accessed within the black box device. In a further embodiment, answers to a query submitted to the black box device by a third party may only be provided with respect to the user data of users who have approved, either in advance or at the time, the use of their user data with respect to such queries.

[0043] Moreover, users may be compensated for the use of their data. Users may provide direct access to their user data in exchange for a monetary fee, a discount on associated products/services, and/or other forms of compensation. Users may, in addition to or instead of providing direct access to their user data, allow their user data to be used in responding to the queries submitted to the black box device as previously described. Such usage may be similarly compensated. Users may grant permission in response to specific bids or requests by third parties according to any one or more of the third party itself, the use to which the user data will be put, and the offered compensation. Users may set parameters in advance (for example, a minimum amount of compensation and/or accepting all but certain excluded uses). Users may respond to specific bids or requests. Users may interact with the system disclosed herein through a combination of preset parameters and dynamic responses to specific bids or requests.

[0044] In an embodiment, because the user data is never directly accessed, only the result of the query about the user data, the present invention advantageously provides an improvement in privacy over systems that permit direct access to the user data and/or permit copying thereof. Data interrogations are accordingly advantageously limited to those entities that have permission to use the user data, entities that have paid to use the data, and/or entities that ask sufficiently broad questions of the user data that anonymity is preserved.

[0045] The black box device is advantageously a secure data store for stored user data. Such user data may include voluntarily supplied information, information gathered from online activities, purchase patterns, movements, etc. The black box device may include a database for storing the user data. The black box device **18** may also include a permissions database that includes specific actions associated with each data type with respect to the user. These actions may be any one or more of elements like ‘never share’, ‘share only with family’, ‘share only for 1 week’, ‘share only with authorized government requests’, etc.

[0046] Accordingly, sharing the user data and/or providing access thereto is conditional on a per item basis and may further depend on the nature and/or identity of an end user inquiring as well as reasons provided by the end user. An additional filter blocks requests that are not sufficiently anonymized.

[0047] The systems, methods, servers, and devices described herein have physical existence and/or manifest a discernible physical effect or change. The systems, methods, servers, and devices described herein relate to the manual or productive arts, meaning those arts involving or concerned with applied and industrial sciences. The computer systems described herein have a material effect on the working of the invention and cooperate with other elements of the claimed invention.

[0048] Where the computer systems herein are programmed to run an algorithm, the computer processes the algorithm in a novel and inventive manner and the processing of the algorithm on the computer solves problems in the functioning of the computer. The computer and the algorithm form part of a single actual invention that solves a problem related to the manual or productive arts. Running the algorithms described herein on the computer improves the functioning of the computer, and the computer and the algorithm together form a single actual invention that solves a problem related to the manual or productive arts.

[0049] The functional improvement particular to the systems, methods, servers, and devices described herein includes but is not limited to saving backend processing and foreground display

through providing the functionality of limiting access to stored user data via the black box device. Limiting access to the stored user data may advantageously avoid unnecessary transmission of large quantities of user data to many end users; because only data approved to be transmitted by the associated user and/or excluded query result data are so transmitted, the present systems, methods, devices, and servers advantageously perform fewer computer operations and/or less computer processing, thereby resulting in an increase in computer efficiency, compared to systems, methods, devices, and servers that lack the present novel and inventive functionality.

[0050] While the present disclosure describes the invention in the context of data integrity validation, the systems, methods, and devices provided herein may have further applications and different uses beyond those described herein, whether in the context of user privacy or otherwise (e.g., cybersecurity, financial transactions, conflicts of interest in professional firms).

[0051] Referring now to FIG. 1, shown therein is a data integrity validation system **10**, in accordance with an embodiment. The system **10** includes a database server **12** for storing user data, which communicates with a network **14**, a user device **16** for capturing and transmitting the user data to the database server **12**, a black box device **18** for locally answering queries received from an end user device without revealing the stored user data, and an end user device **20** for transmitting the queries to the black box device **18**.

[0052] The database server **12** receives the user data through the user device **16**. For example, the user device **16** may receive user data from a user of the user device **16** to be transmitted to the database server **12**. The user data may deliberately be provided by the user (who may wish to provide their user data to the database server **12** in order to gain access to the benefits therefrom herein disclosed). The provision of the user data may further be incidental on the part of the user in the course of accessing other goods and services. Such other goods and services may be affiliated with the system **10**. Such other goods and services may not be affiliated with the system **10**.

[0053] The database server **12** receives the user data through the network **14**. In an embodiment, the network **14** is the Internet.

[0054] The database server **12** further transmits the user data to the black box device **18**. The database server **12** may automatically transmit all the user data received to the black box device. The database server **12** may only transmit unique user data (for example, not transmitting the same user data multiple times). The database server **12** may only transmit certain user data to the black box device **18**. Such transmission may occur according to filters and/or other parameters configured in advance, periodically updated, or both, as to the relevance and/or value of the user data received at the user device **16**. Such filters and/or other parameters may involve the application of artificial intelligence or machine learning. The database server **12** may be configured to update the user data at the black box device **18**. The database server **12** may be configured to send the user data to the black box device **18** in response to a request from the black box device **18**. The request may be for particular user data, for a particular class of user data, and/or for all the user data. The foregoing modes of transmitting user data from the database server **12** to the black box device **18** are not mutually exclusive, and further modes of transmission may be utilized.

[0055] Other methods for collecting and further creating data for possible transmission through the previously mentioned filters to the black box device **18** include through an application or web service such as a shopping or information portal, through a connection to an affinity program that details purchases and interest by specific persons, through social media channels that log interest in specific products and activities, through questionnaires, through gamified interfaces or web plug-ins, through communications content (e.g., email, voice capture) of individuals, and through geolocation for affiliation with visiting specific places.

[0056] The devices **12**, **16**, **18**, **20** may be a server computer, node computing device, embedded device, desktop computer, notebook computer, tablet, PDA, smartphone, or another computing device. The devices **12**, **16**, **18**, **20** may include a connection with the network **14** such as a wired or wireless connection to the Internet. In some cases, the network **14** may include other types of

computer or telecommunication networks. The devices **12, 16, 18, 20** may include one or more of a memory, a secondary storage device, a processor, an input device, a display device, and an output device. Memory may include random access memory (RAM) or similar types of memory. Also, memory may store one or more applications for execution by a processor. Applications may correspond with software modules comprising computer executable instructions to perform processing for the functions described below. The secondary storage device may include a hard disk drive, floppy disk drive, CD drive, DVD drive, Blu-ray drive, or other types of non-volatile data storage. The processor may execute applications, computer readable instructions, or programs. The applications, computer readable instructions, or programs may be stored in memory or in secondary storage or may be received from the Internet or other network **14**.

[0057] The input device may include any device for entering information into the devices **12, 16**. For example, the input device may be a keyboard, keypad, cursor-control device, touchscreen, camera, or microphone. The display device may include any type of device for presenting visual information. For example, the display device may be a computer monitor, a flat-screen display, a projector, or a display panel. The output device may include any type of device for presenting a hard copy of information, such as a printer, for example. The output device may also include other types of output devices such as speakers, for example. In some cases, the devices **12, 16, 18** may include multiple of any one or more of processors, applications, software modules, secondary storage devices, network connections, input devices, output devices, and display devices.

[0058] Although devices **12, 16, 18, 20** are described with various components, one skilled in the art will appreciate that the devices **12, 16, 18, 20** may in some cases contain fewer, additional, or different components. In addition, although aspects of an implementation of the devices **12, 16, 18, 20** may be described as being stored in memory, one skilled in the art will appreciate that these aspects can also be stored on or read from other types of computer program products or computer-readable media, such as secondary storage devices, including hard disks, floppy disks, CDs, or DVDs; a carrier wave from the Internet or other network; or other forms of RAM or ROM. The computer-readable media may include instructions for controlling the devices **12, 16, 18, 20** and/or processor to perform a particular method.

[0059] The devices **12, 16, 18, 20** can be described performing certain acts. It will be appreciated that any one or more of these devices may perform an act automatically or in response to an interaction by a user of that device. That is, the user of the device may manipulate one or more input devices (e.g., a touchscreen, a mouse, a button) causing the device to perform the described act. In many cases, this aspect may not be described below but will be understood.

[0060] As an example, it is described below that the devices **12, 16, 18, 20** may send information to one or more other device **12, 16, 18, 20**. For example, a user using the user device **16** may manipulate one or more inputs (e.g., a mouse and a keyboard) to interact with a user interface displayed on a display of the device **16**. Generally, the device may receive a user interface from a network (e.g., in the form of a webpage). Alternatively, or in addition, a user interface may be stored locally at a device (e.g., a cache of a webpage, a mobile application).

[0061] The devices **12, 16, 18, 20** may be configured to receive a plurality of information, from another one or more of the plurality of devices **12, 16, 18, 20**.

[0062] In response to receiving information, the respective device **12, 16, 18, 20** may store the information in a storage database. The storage may correspond with secondary storage of one or more other devices **12, 16, 18, 20**. Generally, the storage database may be any suitable storage device such as a hard disk drive, a solid-state drive, a memory card, or a disk (e.g., CD, DVD, Blu-ray). Also, the storage database may be locally connected with the device **12, 16, 18, 20**. In some cases, the storage database may be located remotely from the device **12, 16, 18, 20** and accessible to the device **12, 16, 18, 20** across a network, for example. In some cases, the storage database may comprise one or more storage devices located at a networked cloud storage provider.

[0063] The database server **12** may be a purpose-built machine designed specifically for storing the

user data and other related information received through the network **14** and/or the user device **16**. [0064] The network **14** captures the user data. The user data may include information pertaining to a single user or to a variety of users. The information may be voluntarily provided by the users about themselves or about others. The information may also be collected based on available user data, such as websites visited, keywords searched, products or services purchased, and electronic social interactions. The information may be a combination of information voluntarily provided and information collected according to the foregoing. The network **14** may be configured to capture all available user data and to forward all the available user data directly to the database server **12**.

Alternately, the network **14** and/or the database server **12** may be configured with filters, parameters, artificial intelligence, or machine learning capabilities to only forward certain user data based on relevance, value, etc. Such filters, parameters, etc. may be hardcoded upon initialization, preset by users and/or end users, and/or dynamically updated by users and/or end users.

[0065] The user device **16** includes a user interface component (or module) (e.g., a human-machine interface). The user device **16** is configured to capture the user data. The user data may include information pertaining to a single user or to a variety of users. The information may be voluntarily provided by the users about themselves or about others. The information may also be collected based on available user data, such as websites visited, keywords searched, products or services purchased, and electronic social interactions. The user device **16** may be configured to capture all available user data and to forward all the available user data directly to the database server **12**.

Alternately, the user device **16** and/or the database server **12** may be configured with filters, parameters, artificial intelligence, or machine learning capabilities to only forward certain user data based on relevance, value, etc. Such filters, parameters, etc. may be hardcoded upon initialization, preset by users and/or end users, and/or dynamically updated by users and/or end users.

[0066] The user interface component of the user device **16** may also render one or more user interface elements for receiving direct input from the user. For example, the user interface component may provide a yes/no or similar binary option for receiving user input data indicating a selection of an option. In an embodiment, the database server **12** may be configured to ask questions to or perform a survey on the user and store the associated answers or survey data.

[0067] The black box device **18** receives the user data from the database server **12**. The black box device **18** further receives queries from an end user device **20**. The end user device **20** may be configured for use by, for example, marketing companies or advertisers.

[0068] In an embodiment, the black box device **18** includes a memory (not shown) for storing the user data, data filters, filtered data, the queries received from the end user device, results of the queries, and excluded query result data. The black box device **18** further includes a processor (not shown) in communication with the memory and configured to generate the filtered data based on the stored user data and the data filters. The processor is further configured to determine whether each query received that addresses the filtered data is permitted based on properties of the query. The processor is further configured to determine whether the query, if answered locally, would produce a result that would facilitate identifying any particular user. The processor is further configured to provide a response to each permitted query based on the generated filtered data. The processor is further configured to generate the excluded query result data by excluding non-permitted stored user data and stored user data that would facilitate identifying any particular user from the response to each permitted query. The processor is further configured to transmit the excluded query result data to the end user device.

[0069] All information received by the black box device **18** from the database server **12**, including the user data, remains local to the black box device **18**. Such information is not further forwarded or transmitted to the end user device **20** and cannot leave the black box device **18**. Accordingly, when the end user device **20** sends a query to the black box device **18**, the black box device **18** conducts analysis of the user data internally and answers the query locally. That answer, without any of the user data, is then transmitted back to the end user device **20**. In this fashion, the user data

originally transmitted by the network **14** and/or by the user device **16** remains confidential and hidden with respect to the end user device **20**. Accordingly, the black box device **18** may be considered to generate other data and/or metadata with respect to the user data, and only the other data and/or the metadata is transmissible from the black box device **18** to the end user device **20**. [0070] In an embodiment, the end user device **20** is configured to receive the queries from an end user, transmit the queries to the black box device, and display the excluded query result data to the end user.

[0071] In an embodiment, the end user device **20** is configured to ask questions as to the geographical locations or socioeconomic strata of users who may be interested in the end user's products or services. Such users may be identified according to online interaction with the end user or recent purchases of the end user's products or services. As an example, the end user may query through the end user device **20** the average yearly income of all implicated end users or the cities in which the largest clusters of implicated end users may be found. The black box device **18** may receive this query and may review the data received from the database server **12**. The black box device **18** may have already received the data or may request transmission or updating of the data from the database server **12**. The black box device **18** may request further data from the database server **12** to respond to the query. The black box device **18** calculates an answer to the query and transmits the answer back to the end user device **20** without revealing personal or identifying information of the users. For example, the black box device **18** may respond that the average yearly income queried is \$60,000 without revealing the income of any users. As a further example, the black box device **18** may respond that the largest clusters of users implicated are in Toronto, Montreal, and Vancouver, without revealing the actual location of any particular user.

[0072] The end user may be able to specify further parameters of the data to be considered within the black box device **18** in answering the query. For example, such data may be limited to data received or updated within the last 12 months.

[0073] In another embodiment, the end user device **20** may provide information to the black box device **18** as to the identity of the end user and/or the proposed use of the user data, as well as the timing of queries. Such information may then be transmitted to the user through the network **14** and/or the user device **16**. The user corresponding to the user data may be able to set permissions in advance or grant permission for their data to be used by the black box device **18** to answer the query at the time of answering the query or in advance thereof. These permissions may take the form of a permissions matrix.

[0074] In an embodiment, the permissions matrix stores circumstances under which the user allows the user data to be used to respond to the queries, and the user data is not used to respond to the queries unless circumstances provided by the end user device **20** are included in the permissions matrix.

[0075] Permissions granted by the user may be compared with the information provided by the end user to prevent unauthorized use and produce an audit trail for the owner of the user data. The audit trail may include a metadata log. The permissions may be dynamically updated such that the owner of the user data may authorize only specific uses of the user data once a request by an end user is made. For example, the permissions may include a permission item that prevents marketing research from using the user data but allows charitable organizations to do so.

[0076] Upon comparison of permissions, the end user may either be granted or denied the use of the data in generating the query result.

[0077] The system **10**, specifically the black box device **18**, may supervise the use to which the user data is put to confirm compliance both with the parameters of the end user's initial query and with the permissions specified by the user.

[0078] The storing of user-granted permissions and/or the comparison(s) therewith may take place on the black box device **18** or elsewhere within the system **10**. The permission matrix may be stored in a database format with data elements identified by type. For example, a data type may be

“Annual Income” with a numerical value for a particular person of \$80,000. The permission matrix may have a corresponding element with a field value, for example any one of “Do not disclose”, “Disclose only to charities”, “Disclose only to registered banks”, “Disclose only for fee of \$1 or more”. An inquiry generates a response according to both the data value and the response type. In an embodiments, the field values include logical expressions produced through IF, AND, OR, and/or NOT operators, for example, “Disclose to charities” AND NOT “Disclose to Municipal Government”, or “Disclose to bank” IF “In USA”.

[0079] In another embodiment, access to user data is provided in exchange for payment by the end user, which is then remitted entirely or in part to the user. For example, the user may specify a minimum dollar amount for direct access to their user data or a different, perhaps lesser, amount to include their user data in the non-identifying excluded query result data. These prices and whether the end user has paid these prices may form a kind of data filter. For example, a user may assign elements to their “Annual Income” data of “Disclose only for fee of \$10 or more” IF “non-anonymous disclosure” and “Disclose only for fee of \$1 or more” IF “anonymous disclosure”. In the foregoing example, the user has previously indicated permission to disclose their precise annual income if paid at least \$10 for that data and permission to disclose their annual income in an anonymous fashion (e.g., as an answer to the question “What is the average yearly income of all people who have purchased my product at least once?”) if paid at least \$1. It will be appreciated that further variations of fully or partially anonymous or non-anonymous disclosure are expressly contemplated, e.g., agreeing to the disclosure of one's income to respond to the foregoing question as part of a median rather than a mean average such that the end user would see a list of all incomes of all people who have purchased the product at least once and thus see the precise income of the user but not learn the user's identity. In such an example, the user may further specify permissions of “Disclose only for fee of \$1 or more” IF “anonymous disclosure” AND “aggregation of data” (e.g., to generate a mean average), “Disclose only for fee of \$5 or more” IF “anonymous disclosure” AND “non-aggregation of data” (e.g., to generate a median average such that the user's precise datum but not their identity will be provided to the end user), and “Disclose only for fee of \$10 or more” IF “non-anonymous disclosure” (e.g., to disclose both the user's precise datum and their identity to the end user).

[0080] The minimum dollar amounts may be fixed or variable in respect of the user data or otherwise. For example, the user may specify a minimum dollar amount for access to any datum, may charge different amounts for different data, and/or may charge different amounts depending on the end user or the proposed use of the data.

[0081] Accordingly, in an embodiment, the black box device **18** further includes a minimum bid amount for which the user allows the user data to be used to respond to the queries. The user data is not used to respond to the queries unless a bid by the end user meets or exceeds the minimum bid amount.

[0082] The price of an inquiry is thus calculated and the end user is advised as to the cost of ‘running’ such an inquiry. Upon payment of the fee, an inquiry is run and the result delivered to the end user in respect of only those users whose minimum bid amount has been paid by the end user.

[0083] Fees may be set based on the specificity of the inquiry in groups or buckets. For example, a widespread query for all Canadians may have a very small fee per user data point, whereas an inquiry that specifies a city or neighbourhood may command a higher fee.

[0084] In an embodiment, the price is a market or “floating” price, and the price is set by an algorithm designed to set prices for maximum market activity. Thus, prices are set to a level that ensures the best compromise between end user and data owner (user). Such a market pricing scheme is advantageously sensitive to numbers of buyers and sellers, generality of the queries, historic pricing, and demand for data.

[0085] The user may specify minimum dollar amounts in advance to be updated from time to time and/or dynamically. The user may answer particular end user requests for the user data through the

network **14** and/or the user device **16** via the black box device **18** and the end user device **20**. In an embodiment, requests for access to the user data (and excluded query result data based thereon) are automatically denied unless the user accepts a bid for information from the end user.

[0086] An end user may make a general query for information and an associated bid and may specify data to which access is to be purchased (temporarily or otherwise), after which time the bidding ceases.

[0087] The system **10** may be configured to maximize user revenue (e.g., by not accepting any bids less than the associated minimum dollar amount but automatically accepting all bids greater than the associated dollar amount), to minimize end user expenditure (e.g., where an end user overbids, the system **10** only charges the end user the associated minimum dollar amount), or otherwise. The system **10** may charge a fee for these and other services.

[0088] The storing of minimum dollar amounts associated with the user data and/or the comparison(s) of end user bids therewith may take place on the black box device **18** or elsewhere within the system **10**.

[0089] Referring now to FIG. 2, shown therein is a block diagram of a computing device **1000** of the system **10** of FIG. 1, according to an embodiment. The computing device **1000** may be, for example, any one of devices **12**, **16**, **18** of FIG. 1.

[0090] The computing device **1000** includes multiple components such as a processor **1020** that controls the operations of the computing device **1000**. Communication functions, including data communications, voice communications, or both may be performed through a communication subsystem **1040**. Data received by the computing device **1000** may be decompressed and decrypted by a decoder **1060**. The communication subsystem **1040** may receive messages from and send messages to a wireless network **1500**.

[0091] The wireless network **1500** may be any type of wireless network, including, but not limited to, data-centric wireless networks, voice-centric wireless networks, and dual-mode networks that support both voice and data communications.

[0092] The computing device **1000** may be a battery-powered device and as shown includes a battery interface **1420** for receiving one or more rechargeable batteries **1440**.

[0093] The processor **1020** also interacts with additional subsystems such as a Random Access Memory (RAM) **1080**, a flash memory **1110**, a display **1120** (e.g., with a touch-sensitive overlay **1140** connected to an electronic controller **1160** that together comprise a touch-sensitive display **1180**), an actuator assembly **1200**, one or more optional force sensors **1220**, an auxiliary input/output (I/O) subsystem **1240**, a data port **1260**, a speaker **1280**, a microphone **1300**, short-range communications systems **1320** and other device subsystems **1340**.

[0094] In some embodiments, user-interaction with the graphical user interface may be performed through the touch-sensitive overlay **1140**. The processor **1020** may interact with the touch-sensitive overlay **1140** via the electronic controller **1160**. Information, such as text, characters, symbols, images, icons, and other items that may be displayed or rendered on a computing device generated by the processor **1020** may be displayed on the touch-sensitive display **1180**.

[0095] The processor **1020** may also interact with an accelerometer **1360**. The accelerometer **1360** may be utilized for detecting direction of gravitational forces or gravity-induced reaction forces.

[0096] To identify a subscriber for network access according to the present embodiment, the computing device **1000** may use a Subscriber Identity Module or a Removable User Identity Module (SIM/RUIM) card **1380** inserted into a SIM/RUIM interface **1400** for communication with a network (such as the wireless network **1500**). Alternatively, user identification information may be programmed into the flash memory **1110** or performed using other techniques.

[0097] The computing device **1000** also includes an operating system **1460** and software components **1480** that are executed by the processor **1020** and which may be stored in a persistent data storage device such as the flash memory **1110**. Additional applications may be loaded onto the computing device **1000** through the wireless network **1500**, the auxiliary I/O subsystem **1240**, the

data port **1260**, the short-range communications subsystem **1320**, or any other suitable device subsystem **1340**.

[0098] In use, a received signal such as a text message, an e-mail message, web page download, or other data may be processed by the communication subsystem **1040** and input to the processor **1020**. The processor **1020** then processes the received signal for output to the display **1120** or alternatively to the auxiliary I/O subsystem **1240**. A subscriber may also compose data items, such as e-mail messages, for example, which may be transmitted over the wireless network **1500** through the communication subsystem **1040**.

[0099] For voice communications, the overall operation of the computing device **1000** may be similar. The speaker **1280** may output audible information converted from electrical signals, and the microphone **1300** may convert audible information into electrical signals for processing.

[0100] Referring now to FIG. 3, shown therein is a block diagram of a system **300** for implementing data integrity validation.

[0101] The system **300** includes a processor **302**, a memory **304**, a communication interface **306** for interacting with the end user, and a display **308** for displaying the results of a query to the end user.

[0102] The memory **304** includes user data **320**. The user data **320** may include any or all of the user data of the database server **12** of FIG. 1.

[0103] The memory **304** further includes data filters **322** for filtering the user data **320** on the basis of relevance, interest, value, etc. The data filters **322** may be hardcoded, preset by the user and/or end user, and/or dynamically updated by the user and/or end user.

[0104] The processor **302** includes a data filtering module **312** for applying the data filters **322** to the user data **320** to generate filtered data **324**. The filtered data **324** is stored in the memory **304**.

[0105] The memory **304** further includes a query **326** received within the system **300**. The query **326** is generated and transmitted, for example by the end user device **20**, through the communication interface **306**.

[0106] The processor **302** includes a query analysis module **314** for analyzing the query **326** and determining what filtered data **324** will satisfy the query **326**. This analysis may be performed through NLP, AI, machine learning, etc. This analysis includes determining whether the end user query **326** is permitted. This analysis includes determining whether the end user query **326**, if answered locally, would produce a result that would facilitate identifying any particular user.

[0107] In an embodiment, the permission matrix is stored in a database format with data elements identified by type. For example, a data type may be “Annual Income” with a numerical value for a particular person of \$80,000. The permission matrix may have a corresponding element with a field value, for example any one of “Do not disclose”, “Disclose only to charities”, “Disclose only to registered banks”, “Disclose only for fee of \$1 or more”. An inquiry generates a response according to both the data value and the field values. In an embodiment, the field values include logical expressions produced through IF, AND, OR, and/or NOT operators, for example, “Disclose to charities” AND NOT “Disclose to Municipal Government”, or “Disclose to bank” IF “In USA”.

[0108] The processor **302** further includes a query response module **316** for responding to the query **326** on the basis of the result of the query analysis module **314** and the filtered data **324**. The query response module **316** generates a result for the query, which is stored in the memory **304** at query result **328**.

[0109] The processor **302** further includes a query exclusion module **318** for excluding specific, user-identifying data from the query result **328**. This exclusion may be based on general principles concerning digital anonymity, on applicable privacy legislation, and/or on the data filters **322** as hardcoded, preset, and/or dynamically updated within the system **300**. The exclusion is configured according to the limits that a user has placed on their privacy to set the limit to which the query **326** can identify a specific individual. The query result **328** is analyzed to determine how many individuals have a positive result to the query **326**. The number of individuals with the positive

result to the query **326** is compared to a personal identification threshold. For example, if **3** individuals were identified as relevant or responsive for the query **326**, the query **326** may be sufficiently narrow to block the query results **328** according to the limits that the user has placed on their privacy. If the query **326** has 1000 individuals identified as relevant or responsive, the user is sufficiently anonymized to give permission to use the user data. The user may be able to select the desired threshold to preserve anonymity.

[0110] In addition, thresholds may be based on privacy legislation, for example not being able to determine the location of an individual with greater granularity than a district or city block.

[0111] The result of processing by the query exclusion module **318** on the query result **328** is stored in the memory **304** as excluded query result data **330**.

[0112] The excluded query result data **330** is provided to the end user, either through display at the display **308** or otherwise. Accordingly, the end user is able to see the excluded query result data **330**, i.e., the query result **328** after inappropriately identifying information has been removed therefrom. Information excluded from the query result **328** is not provided to the end user.

[0113] Advantageously, the foregoing embodiment prevents the transmission of any specific, user-identifying data of any users to the end user. Accordingly, the end user is able to receive answers to specific queries generated by detailed analysis on the user data without the anonymity or privacy of the users being compromised.

[0114] Referring now to FIG. **4**, shown therein is a method **400** of using user data and data filters to respond to an end user query.

[0115] At **402**, user data **320** is received, for example from the network **14** or the user device **16**.

[0116] At **404**, data filters **322** are received.

[0117] At **406**, the data filters **322** are applied to the user data **320** to generate filtered data **324**.

[0118] At **408**, an end user query **326** is received, for example from the end user device **20**.

[0119] At **410**, the end user query **326** is analyzed at the query analysis module **314** to determine that the end user query **326** is permitted.

[0120] At **411**, the end user query **326** is analyzed at the query analysis module **314** to determine that the end user query **326**, if answered locally, would not produce a result that would facilitate identifying any particular user.

[0121] At **412**, a query result **328** is generated at the query response module **318**. Such generation may take place according to the filtered data **324**.

[0122] At **414**, non-permitted user data and user data that would facilitate identifying any particular user are excluded from the query result **328** to generate excluded query result data **330**.

[0123] At **416**, the excluded query result data **330** is transmitted to the end user device **20**.

[0124] The method **400** may include storing circumstances under which the user allows the user data to be used to respond to the query in a permissions matrix and not allowing the user data to be used to respond to the query unless circumstances provided by the end user are included in the permissions matrix.

[0125] The method **400** may include allowing the user data to be used to respond to the queries when a bid by the end user exceeds a minimum bid amount and not allowing the user data to be used to respond to the query if the bid by the end user does not meet or exceed the minimum bid amount.

[0126] Referring now to FIG. **5**, shown therein is a method **500** for determining access to user data according to an end user request and based on permissions.

[0127] At **502**, the end user request for access to the user data is received.

[0128] At **504**, details on the end user and the request are received, such as the end user identity and the proposed use of the user data.

[0129] At **506**, the user permissions for the user data are updated. This may include querying the user for new or current user permissions.

[0130] At **508**, permissions granted by the user are compared with the details provided by the end

user.

[0131] At **510**, access to the user data is granted based on the comparison if the permissions match, i.e., if the end user details are included within the scope of the user permissions.

[0132] At **512**, access to the user data is denied based on the comparison if the permissions do not match.

[0133] At **514**, the use to which the user data is put is supervised.

[0134] Turning now to FIG. **6**, shown therein is a method **600** for determining access to user data according to an end user request and based on bidding.

[0135] At **602**, the end user request for access to the user data is received.

[0136] At **604**, a financial bid by the end user for the user data is received.

[0137] At **606**, a minimum dollar amount associated with the user data is updated from the user.

[0138] At **608**, the end user bid and the user minimum dollar amount are compared.

[0139] At **610**, access to the user data is granted based on the comparison if the end user bid meets or exceeds the minimum dollar amount.

[0140] At **612**, access to the user data is denied based on the comparison if the end user bid does not meet or exceed the minimum dollar amount.

[0141] At **614**, at least partial payment is remitted to the user according to the end user bid.

[0142] It will be appreciated that, in an embodiment, the methods **500** and **600** are combined such that access to the user data is granted or denied based on a comparison of the permissions (i.e., whether the end user details are included within the scope of the user permissions) and a comparison of the end user bid and the user-provided minimum dollar amount.

[0143] It will be appreciated that, in an embodiment, the method **500**, the method **600**, or the combined method according to the methods **500** and **600** may further include all or part of the method **400** to be performed upon access to the user data being granted.

[0144] It will be appreciated that, in an embodiment, **410** of the method **400** may include performing one or both of all or part of the methods **500** and/or **600**.

[0145] It will be appreciated that, in an embodiment, **411** of the method **400** may include performing one or both of all or part of the methods **500** and/or **600**.

[0146] While the above description provides examples of one or more apparatus, methods, or systems, it will be appreciated that other apparatus, methods, or systems may be within the scope of the claims as interpreted by one of skill in the art.

Claims

1. A system for data integrity validation, the system comprising: a user database for storing user data of a user; a black box device for locally answering queries received from an end user device without revealing the stored user data, the black box device comprising: a memory for storing the user data, data filters, filtered data, the queries received from the end user device, results of the queries, and excluded query result data; and a processor in communication with the memory, the processor configured to: generate the filtered data based on the stored user data and the data filters; determine whether each query received that addresses the filtered data is permitted based on properties of the query; determine whether the query, if answered locally, would produce a result that would facilitate identifying any particular user; provide a response to each permitted query based on the generated filtered data; generate the excluded query result data by excluding from the response to each permitted query non-permitted stored user data and stored user data that would facilitate identifying any particular user; and transmit the excluded query result data to the end user device; the end user device configured to: receive the queries from an end user; transmit the queries to the black box device; and display the excluded query result data to the end user.

2. The system of claim 1, wherein the black box device further comprises a permissions matrix for storing circumstances under which the user allows the user data to be used to respond to the queries

and wherein the user data is not used to respond to the queries unless circumstances provided by the end user device are included in the permissions matrix.

3. The system of claim 2, wherein the permissions matrix notifies the user when the circumstances provided by the end user device are not included in the permissions matrix, and wherein the permissions matrix is configured to receive permission from the user to allow the user data to be used to respond to the queries when the circumstances provided by the end user device are included in the permissions matrix.

4. The system of claim 2, wherein use by the end user of the excluded query result data is audited to confirm that the use by the end user matches the circumstances provided by the end user device.

5. The system of claim 1, wherein the black box device further comprises a minimum bid amount for which the user allows the user data to be used to respond to the queries and wherein the user data is not used to respond to the queries unless a bid by the end user device meets or exceeds the minimum bid amount.

6. The system of claim 5, wherein the minimum bid amount includes a first minimum bid amount for anonymous disclosure of the user data, a second minimum bid amount for semi-anonymous disclosure of the user data, the second minimum bid amount being higher than the first minimum bid amount, and a third minimum bid amount for non-anonymous disclosure of the user data, the third minimum bid amount being higher than the second minimum bid amount.

7. The system of claim 1 further comprising a user device for providing the user data.

8. The system of claim 1, wherein the user data is gathered from any one or more of online activities of the user, purchase patterns of the user, and physical movement of the user.

9. A method for facilitating data integrity validation, the method comprising: receiving user data pertaining to one or more users; receiving data filters to be applied to the user data; applying the data filters to the user data to generate filtered data; receiving a query from an end user device; determining that the query received is permitted based on properties of the query; determining that the query, if answered locally, would not produce a result that would facilitate identifying any particular user; providing a response to the permitted query using the filtered data; generating excluded query result data by excluding from the response to each permitted query non-permitted user data and user data that would facilitate identifying any particular user; and transmitting the excluded query result data to the end user.

10. The method of claim 9 further comprising storing circumstances under which the user allows the user data to be used to respond to the query in a permissions matrix and not allowing the user data to be used to respond to the query unless circumstances provided by the end user are included in the permissions matrix.

11. The method of claim 10, further comprising: the permissions matrix notifying the user when the circumstances provided by the end user device are not included in the permissions matrix; and the permissions matrix receiving permission from the user to allow the user data to be used to respond to the queries when the circumstances provided by the end user device are included in the permissions matrix.

12. The method of claim 10 further comprising auditing use by the end user of the excluded query result data to confirm that the use by the end user matches the circumstances provided by the end user device.

13. The method of claim 9 further comprising allowing the user data to be used to respond to the queries when a bid by the end user device meets or exceeds a minimum bid amount and not allowing the user data to be used to respond to the query if the bid by the end user does not meet or exceed the minimum bid amount.

14. The method of claim 13, wherein the minimum bid amount includes a first minimum bid amount for anonymous disclosure of the user data, a second minimum bid amount for semi-anonymous disclosure of the user data, the second minimum bid amount being higher than the first minimum bid amount, and a third minimum bid amount for non-anonymous disclosure of the user

data, the third minimum bid amount being higher than the second minimum bid amount.

15. A black box device for locally answering queries received from an end user device without revealing stored user data, the black box device comprising: a memory for storing the user data of a user, data filters, filtered data, the queries received from the end user device, results of the queries, and excluded query result data; and a processor in communication with the memory, the processor configured to: generate the filtered data based on the stored user data and the data filters; determine whether each query received that addresses the filtered data is permitted based on properties of the query; determine whether the query, if answered locally, would produce a result that would facilitate identifying any particular user; provide a response to each permitted query based on the generated filtered data; generate the excluded query result data by excluding from the response to each permitted query non-permitted stored user data and stored user data that would facilitate identifying any particular user; and transmit the excluded query result data to the end user device.

16. The device of claim 15 further comprising a permissions matrix for storing circumstances under which the user allows the user data to be used to respond to the queries, wherein the user data is not used to respond to the queries unless circumstances provided by the end user are included in the permissions matrix.

17. The device of claim 16, wherein the permissions matrix notifies the user when the circumstances provided by the end user device are not included in the permissions matrix, and wherein the permissions matrix is configured to receive permission from the user to allow the user data to be used to respond to the queries when the circumstances provided by the end user device are included in the permissions matrix.

18. The device of claim 16, wherein use by the end user of the excluded query result data is audited to confirm that the use by the end user matches the circumstances provided by the end user device.

19. The device of claim 15 further comprising a minimum bid amount for which the user allows the user data to be used to respond to the queries, wherein the user data is not used to respond to the queries unless a bid by the end user device meets or exceeds the minimum bid amount.

20. The device of claim 19, wherein the minimum bid amount includes a first minimum bid amount for anonymous disclosure of the user data, a second minimum bid amount for semi-anonymous disclosure of the user data, the second minimum bid amount being higher than the first minimum bid amount, and a third minimum bid amount for non-anonymous disclosure of the user data, the third minimum bid amount being higher than the second minimum bid amount.
